

Highlights

Stochastic Physical-Ensemble Quantiles Improve Hybrid Machine-Learning Simulation of Daily Streamflow in the High-Andean Ramis Basin

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- Ensemble quantiles were the dominant driver of hybrid streamflow skill.
- The quantile hybrid improved NSE by 0.311 over stochastic RAMIS.
- Lévy perturbations informed ML features but underdispersed intervals.
- Backend and leakage audits constrained claims for GRU-labelled runs.
- The ablation separated memory, stochasticity, ML and descriptors.

Stochastic Physical-Ensemble Quantiles Improve Hybrid Machine-Learning Simulation of Daily Streamflow in the High-Andean Ramis Basin

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ABSTRACT

Hybrid hydrological modelling combines process-based structure with the predictive flexibility of machine learning (ML), but often leaves unclear which component drives improved streamflow prediction. We present StoHyMoLAP, a reproducible and audit-controlled framework for daily streamflow simulation in the high-Andean Ramis basin, within the Lake Titicaca system of southern Peru. The framework couples a lumped RAMIS rainfall-runoff core, an optional linear baseflow reservoir, Lévy-type (α -stable) stochastic perturbations, Monte Carlo physical ensembles, and ML emulators trained from observed forcings or from stochastic-ensemble descriptors. Nine experiments (E0–E8) were evaluated over a common validation window from 8 January 2009 to 31 December 2020 ($n = 4367$ daily samples), with temporal-leakage and backend audits embedded in the evidence chain.

Quantile descriptors from the stochastic physical ensemble provided the largest robust gain. The gradient-boosting quantile hybrid reached NSE = 0.740 and KGE = 0.810, improving NSE by 0.311 relative to the complete stochastic physical model and by 0.165 relative to the pure ML baseline. Replacing ensemble-mean features with ensemble quantiles alone added 0.114 NSE, showing that distributional physical descriptors contain information not captured by a single mean trajectory. The baseflow reservoir modestly improved the deterministic physical baseline, whereas direct Lévy perturbations produced inconsistent point-prediction gains. The stochastic physical bands were strongly underdispersive (PICP = 0.004 against a nominal 0.95 target), so the ensemble is more useful as ML features than as calibrated predictive uncertainty. The numerically best run (NSE = 0.849, KGE = 0.899, RMSE = $0.210 \text{ m}^3 \text{ s}^{-1}$) was labelled as GRU but executed as a fallback multilayer perceptron on flattened sequences; it is not evidence of recurrent-network superiority. These findings support stochastic physical-ML hybridization while showing that uncertainty calibration, low-flow correction and backend-aware reporting remain essential for defensible hydrological modelling.

1. Introduction

Reliable daily streamflow simulation remains a central problem in surface-water hydrology, particularly in basins where observations are sparse, hydroclimatic variability is strong and operational decisions must still be supported by defensible discharge estimates (Razavi & Coulibaly 2013, Guo, Zhang, Zhang & Wang 2021). These conditions are characteristic of many high-elevation Andean catchments. In the Peruvian Altiplano, precipitation is strongly seasonal, monitoring networks are sparse relative to the spatial heterogeneity of rainfall, and discharge dynamics are shaped by the alternation between sharp wet-season runoff response and sustained dry-season low flows (Condom, Coudrain, Dezetter, Brunstein, Delclaux & Jean-Emmanuel 2004, Zubietta, Laqui & Lavado 2018, Lima-Quispe, Ruelland, Rabatel, Lavado-Casimiro & Condom 2025). Such settings expose a persistent limitation of classical rainfall-runoff modelling: even when a conceptual model captures the main water-balance controls, simplified storage structures, uncertain parameters and imperfect meteorological inputs can lead to structural errors that become particularly visible under low-flow, flashy or climatically contrasting conditions (van Esse, Perrin, Booij, Augustijn, Fenicia, Kavetski & Lobligois 2013, Nauditt, Birkel, Soulsby & Ribbe 2017, Ragettli, Cortés, McPhee & Pellicciotti 2014).

Machine-learning (ML) models offer a complementary route because they can learn nonlinear rainfall-runoff relations directly from data. Large sample studies have shown that recurrent and other data-driven models can

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match or outperform lumped conceptual models in daily streamflow prediction, including in regional and basin-scale benchmarks (Kratzert, Klotz, Shalev, Klambauer, Hochreiter & Nearing 2019, Lees, Buechel, Anderson, Slater, Reece, Coxon & Dadson 2021, Arsenault, Martel, Brunet, Brissette & Mai 2022). However, predictive skill alone does not eliminate hydrological risk. Purely data-driven models may be sensitive to distributional shifts, training-period representativeness, extrapolation to extremes and the absence of explicit physical constraints (Frame, Kratzert, Klotz, Gauch, Shelev, Gilon, Qualls, Gupta & Nearing 2021, Zhong, Lei & Gao 2023). In data-scarce basins, their performance also depends strongly on whether antecedent hydrological state information is available, because rainfall inputs alone may be insufficient to represent catchment memory (Adnan, Petroselli, Heddiam, Santos & Kisi 2021, Okkan, Ersoy, Kumanlioglu & Fistikoglu 2021). The practical question is therefore not whether process-based or ML models are universally superior, but how physical structure and empirical flexibility can be combined without obscuring the source of any improvement.

Hybrid physical–ML modelling addresses this question by using process simulations, physically derived states or synthetic physically consistent samples as inputs to an empirical emulator. This strategy has improved runoff simulation in several configurations, including conceptual-model correction, nested hybrid rainfall–runoff modelling, physics-guided deep learning and process-driven deep learning (Mekonnen, Nazemi, Mazurek, Elshorbagy & Putz 2015, Mohammadi, Safari & Vazifekkhah 2022, Okkan et al. 2021, Yang, Yang, Chen, Santisirisomboon, Lu & Zhao 2020, Xie, Liu, Zhang, Han, Wang & Shen 2021, Li, Zhang, Chu, Shen & Li 2024, Zhu, Zhu, Han, Liu, Guo, Hou, Zhao, Xu & Zhong 2025). Most of these studies, however, pass forward a deterministic physical trajectory. That design preserves part of the process information but collapses the conditional distribution of the simulated response into a single value. For stochastic rainfall–runoff transformation, this is a strong simplification: uncertainty in precipitation, model structure and hydrological memory propagates nonlinearly into discharge and can alter both peak timing and magnitude (Gabellani, Boni, Ferraris, Von Hardenberg & Provenza 2007, Hong, Hsu, Moradkhani & Sorooshian 2006). It also limits the ability of the hybrid model to exploit distributional descriptors such as ensemble quantiles, which are closely related to hydrological signatures and to probabilistic forecasting diagnostics (Guo et al. 2021, Papacharalampous & Tyralis 2022).

A recent stochastic hybridization study by Houénafa, Johnson, Ronoh & Moore (2025) showed that mean and quantile descriptors of a Lévy-driven stochastic HyMoLAP discharge ensemble can improve ML-based rainfall–runoff prediction. This result is important because it moves physical–ML hybridization from deterministic state correction toward distribution-aware physical descriptors. Yet it also leaves three issues unresolved. First, the gain from the hybrid pipeline may come from several components—hydrological memory, stochastic perturbations, the ML correction itself or the use of quantile rather than mean ensemble descriptors—and these effects are not identifiable from a single best-model comparison. Second, a stochastic ensemble that improves point prediction is not necessarily a calibrated predictive interval; coverage, sharpness and interval scores must be evaluated separately. Third, reproducibility claims require that reported model labels match the backend that actually produced the results, especially when deep-learning architectures have software fallbacks or implementation contingencies.

This study addresses these gaps through StoHyMoLAP, an audited stochastic physical–ML framework for daily streamflow simulation in the high-Andean Ramis basin, the principal tributary of Lake Titicaca. The framework couples a lumped RAMIS rainfall–runoff core, an optional linear baseflow reservoir, Lévy-type stochastic perturbations, Monte Carlo physical ensembles and ML emulators trained either from observed forcings or from physical-ensemble descriptors. The purpose is not to introduce another best-performing hybrid model, but to decompose which components are responsible for performance gains. Nine controlled experiments (E0–E8; Table 1) are therefore evaluated over a common validation window using a single workflow (Fig. 1). The ablation design isolates the incremental roles of baseflow memory, stochastic forcing, ML correction and descriptor choice.

The manuscript makes four specific contributions. First, it transfers stochastic physical–ML quantile hybridization to a high-Andean, strongly seasonal and data-scarce Altiplano basin, providing a contrast to previous applications outside this hydroclimatic setting. Second, it shows whether ensemble mean or ensemble quantiles carry the more useful physical information for ML correction. Third, it evaluates the same stochastic ensemble both as a source of predictive features and as a candidate uncertainty band through PICP, PINAW and Winkler diagnostics. Fourth, it embeds temporal-leakage and backend audits into the evidence chain, following the broader call for transparent and reproducible hydrological modelling (Hall, Saia, Popp, Dogulu, Schymanski, Drost, Van Emmerik & Hut 2021). A key audit finding is stated explicitly: the experiments labelled as GRU in the current run were executed with a fallback multilayer perceptron on flattened sequences, not with a true recurrent backend. They are therefore interpreted as

GRU-labelled sequential hybrids under fallback execution, not as evidence of recurrent-network superiority (Kratzert, Klotz, Brenner, Schulz & Herrnegger 2018).

2. Materials and methods

2.1. Study area

The Ramis River basin is located in the high-Andean Altiplano of southern Peru and drains towards Lake Titicaca. The basin is therefore part of a closed lacustrine system in which river inflow, precipitation and evaporation exert strong control on lake-level variability (Cross, Baker, Seltzer, Fritz & Dunbar 2001, Condom et al. 2004, Lima-Quispe et al. 2025). Previous regional studies show that the hydrology of the Titicaca system is controlled mainly by rainfall and evapotranspiration variability, whereas snowmelt, ice melt and net irrigation withdrawals have a secondary role at the lake-basin scale (Lima-Quispe et al. 2025). This regional behaviour makes the Ramis basin a relevant test case for a rainfall–runoff framework designed to combine process-based memory, stochastic forcing and machine-learning correction.

The basin is characterized by high elevation, sparse hydroclimatic monitoring and a strongly seasonal precipitation regime. Rainfall is concentrated during the austral summer wet season, while the dry season is associated with prolonged recession and low-flow persistence (Condom et al. 2004, Nauditt et al. 2017, Lima-Quispe et al. 2025). Such hydroclimatic asymmetry is challenging for lumped conceptual models because a single storage response may be unable to represent both wet-season peaks and dry-season baseflow. It is also challenging for purely data-driven models, which may learn strong seasonal correlations without representing the storage mechanisms that maintain discharge between rainfall events (van Esse et al. 2013, Lees et al. 2021). Daily discharge observations used in this study correspond to the Puente Ramis gauging station operated by SENAMHI-Peru.

2.2. Hydro-meteorological data and temporal protocol

The modelling unit is a daily hydro-meteorological record composed of precipitation (P_t), minimum and maximum temperature ($T_{\min,t}$ and $T_{\max,t}$), potential evapotranspiration (PET_t) and observed streamflow (Q_t^{obs}). The rainfall–runoff core uses an effective precipitation input defined as

$$P_t^{eff} = \max(P_t - PET_t, 0), \quad (1)$$

which provides a non-negative approximation of the water available for runoff generation after atmospheric demand has been accounted for.

All experiments follow the same chronological split. Model calibration and ML training use data from 8 September 1991 to 31 December 2008, whereas validation uses data from 8 January 2009 to 31 December 2020. The final leaderboard is computed over the common validation intersection of all experiments, comprising $n = 4367$ daily samples. This common-window design prevents unequal evaluation support across physical, ML and hybrid configurations and is especially important for experiments that require lagged predictors or sequence construction.

Temporal leakage was controlled at four levels. First, lagged predictors were constructed only from present or past information. Second, sequence samples were generated independently within the training and validation partitions. Third, scalars and transformations used by ML models were fitted exclusively on training data and then applied to validation data. Fourth, flow-regime thresholds used for diagnostic stratification were estimated from training observations only. These controls follow the broader hydrological ML recommendation that model selection and preprocessing must preserve temporal ordering (Arsenault et al. 2022, Chen, Brissette, Chaumont & Braun 2013). The corresponding audit status is reported in Table 7.

2.3. Lumped rainfall–runoff core

Although the implementation workflow occasionally uses the term “hydrodynamic”, the model evaluated here is a lumped stochastic rainfall–runoff simulator, not a spatially distributed hydraulic or hydrodynamic solver. The process-based component is therefore described as a catchment-scale discharge-dynamics model. Its fast-flow state is represented by the recurrence

$$x_t = \left(1 - \frac{\mu}{\lambda}\right) x_{t-1} + P_t^{eff}, \quad (2)$$

where $\mu > 0$ and $\lambda > 0$ regulate catchment memory and recession stability. The parameter constraint $0 < \mu/\lambda < 1$ ensures a stable, non-oscillatory memory factor.

The fast-flow discharge component is updated as

$$Q_t^{fast} = Q_{t-1}^{fast} - \frac{\mu}{\lambda} Q_0 \left(\frac{\max(Q_{t-1}^{fast}, 0)}{Q_0} \right)^{2\mu-1} + \frac{\alpha_A}{\lambda} x_{t-1} + \sigma Q_{t-1}^{fast} \Delta L_t, \quad (3)$$

where Q_0 is a reference discharge, α_A is an empirical conversion factor, σ controls stochastic amplitude and ΔL_t is a stochastic increment. In deterministic experiments, the stochastic term is inactive. Negative simulated discharge is clipped to zero when non-negative physical post-processing is enabled.

The recurrence in Eqs. 2–3 represents catchment-scale nonlinearity through aggregated storage behaviour. This interpretation is consistent with recession analyses in which nonlinear catchment response can emerge from heterogeneous linear storage elements rather than from a single local nonlinear mechanism (Harman, Sivapalan & Kumar 2009). In this setting, the parameter μ acts as a shape parameter controlling the effective recession behaviour of the lumped storage system.

2.4. Linear baseflow reservoir

An optional linear reservoir is used to represent hydrological memory beyond the fast-flow recurrence. This addition is motivated by previous work in the Titicaca region showing that groundwater storage fluxes are important for explaining seasonal runoff persistence and dry-season baseflow (Nauditt et al. 2017). The reservoir also provides a controlled way to test whether adding a simple memory component improves simulation skill before introducing stochastic perturbations or ML correction.

Recharge to the baseflow store is proportional to effective precipitation:

$$R_t = c_r \alpha_A P_t^{eff}, \quad (4)$$

where $c_r \in [0, 1]$ is the recharge coefficient. Storage and baseflow discharge then evolve as

$$S_{b,t+1} = \max(0, S_{b,t} + R_t - k_b S_{b,t}), \quad (5)$$

$$Q_t^{base} = k_b S_{b,t}, \quad (6)$$

where $k_b \in [0.001, 0.5]$ is the linear recession coefficient. The total process-based discharge is

$$Q_t^{phys} = \max(Q_t^{fast} + Q_t^{base}, 0). \quad (7)$$

The baseflow component is activated in E1, E3 and all hybrid configurations. Its isolated contribution is evaluated through the E1–E0 and E3–E2 ablation contrasts.

2.5. Lévy perturbations and Monte Carlo physical ensembles

Stochastic physical experiments inject α -stable Lévy-type increments into Eq. 3. The perturbation is governed by stability, skewness, location and scale parameters. Heavy-tailed increments are used to represent episodic deviations from the smooth lumped response, which is relevant in a highly seasonal basin where rainfall inputs and streamflow errors are not expected to be Gaussian. This design is consistent with hydrological ensemble studies showing that uncertainty is not limited to the mean response but also involves dispersion, asymmetry and tail behaviour (Hong et al. 2006, Gabellani et al. 2007, Abaza, Anctil, Fortin & Perreault 2017).

For each calibrated stochastic configuration, a Monte Carlo ensemble of N trajectories is generated. At each day, the ensemble is summarized by central-tendency, dispersion and quantile descriptors. The quantile set used in the hybrid experiments is

$$\{q_{0.01}, q_{0.05}, q_{0.25}, q_{0.50}, q_{0.75}, q_{0.95}, q_{0.99}\}, \quad (8)$$

together with spread measures such as $q_{0.95} - q_{0.05}$ and $q_{0.75} - q_{0.25}$. Quantile representations are robust summaries of non-Gaussian and fat-tailed distributions, and therefore provide a natural way to pass distributional information from a stochastic physical ensemble to an ML emulator (Swiderski, Osowski, Kruk & Barhoumi 2016). In this study, the same ensemble is evaluated in two roles: as a direct stochastic physical predictor in E2 and E3, and as a source of physically informed ML features in E5–E8.

2.6. Machine-learning emulators and feature sets

The ML component maps a feature vector $\mathbf{z}_t$ to daily simulated discharge,

$$\hat{Q}_t = f_{\theta}(\mathbf{z}_t), \quad (9)$$

followed by non-negative clipping,

$$Q_t^{sim} = \max(\hat{Q}_t, 0). \quad (10)$$

This post-processing step ensures physically admissible discharge predictions while preserving the fitted ML response over positive flows.

Three feature families are compared. The pure ML baseline (E4) uses observed hydro-meteorological forcings and non-negative lags, including lagged runoff information where allowed by the temporal protocol. Including antecedent runoff is hydrologically meaningful because it acts as a proxy for catchment storage and has been shown to improve rainfall–runoff ML models relative to meteorological predictors alone (Adnan et al. 2021). Mean-based hybrids (E5 and E7) use the mean trajectory of the stochastic physical ensemble. Quantile-based hybrids (E6 and E8) use the ensemble quantiles and spread descriptors listed in Eq. 8. This design tests whether the ML emulator benefits primarily from a corrected central physical trajectory or from richer distributional descriptors of the physical ensemble.

The tabular emulator is labelled as XGB in the experiment names. In the current evidence package, however, the actual backend was HistGradientBoostingRegressor from scikit-learn, reported in the audit tables as sklearn_hgb. The corresponding results are therefore interpreted as gradient-boosted tree hybrids rather than as package-specific evidence for xgboost. Hyperparameter search used temporally ordered cross-validation, avoiding random shuffling of time steps during model selection.

The sequential emulator is labelled as GRU in the experiment names and is designed to use recurrent layers when a supported backend is available. In the present execution, the recurrent backend was not available; the implementation flattened each input sequence and trained a multilayer perceptron instead. Consequently, E7 and E8 are retained in the experimental matrix because they evaluate sequentially constructed features, but they are interpreted throughout the manuscript as GRU-labelled fallback-MLP runs, not as evidence of true recurrent GRU superiority (Kratzert et al. 2018, 2019).

2.7. Calibration objective

The process-based and stochastic configurations are calibrated with a scalar objective that combines efficiency, distributional agreement, bias and interval coverage:

$$J = \frac{w_{NSE}(1 - NSE) + w_{KGE}(1 - KGE) + w_{PBIAS} \left| \frac{PBIAS}{100} \right| + w_{cov}|c^* - PICP|}{w_{active}}, \quad (11)$$

where $c^* = 0.95$ is the nominal coverage target and w_{active} is the sum of the active weights. The default weights are $w_{NSE} = 0.40$, $w_{KGE} = 0.30$, $w_{PBIAS} = 0.15$ and $w_{cov} = 0.15$. When a model does not produce prediction intervals, the coverage term is omitted and the remaining weights are renormalized.

The composite objective is used because no single deterministic metric fully describes hydrological model performance, particularly when both high-flow peaks and low-flow persistence are relevant (Pushpalatha, Perrin, Le Moine & Andréassian 2012). The coverage term is included to avoid selecting stochastic configurations that are sharp but severely underdispersive, a common limitation in hydrological ensemble prediction (Franz & Hogue 2011, Abaza et al. 2017).

2.8. Experimental matrix and ablation design

The experimental matrix was designed to isolate the contribution of each modelling component under a common validation window. E0 and E1 compare the deterministic RAMIS core without and with the baseflow reservoir. E2 and E3 add Lévy stochasticity without and with baseflow. E4 is a pure ML baseline trained only from observed predictors. E5 and E6 are gradient-boosting hybrids using, respectively, ensemble-mean and ensemble-quantile physical descriptors. E7 and E8 use the same mean and quantile feature families in a sequential-feature setting, but with the fallback MLP backend described above.

This matrix supports five planned contrasts: the effect of baseflow (E1–E0 and E3–E2), the direct effect of stochastic perturbations under baseflow (E3–E1), the effect of pure ML correction relative to the full stochastic physical model

Table 1

Controlled experimental matrix. Backend entries report the actual backend used in the current evidence package.

Experiment	Family	Lévy	Baseflow	Emulator	Features	Backend used
E0_RAMIS_DET_NOBF	physical RAMIS	no	no	none	physical only	physical
E1_RAMIS_DET_BF	physical RAMIS	no	yes	none	physical only	physical
E2_RAMIS_LEVY_NOBF	RAMIS-Lévy	yes	no	none	physical only	physical
E3_RAMIS_LEVY_BF	RAMIS-Lévy	yes	yes	none	physical only	physical
E4_ML_PURE_XGB	pure ML	no	no	XGB-labelled	observed forcings	sklearn_hgb
E5_HYB_XGB_MEAN	hybrid	yes	yes	XGB-labelled	ensemble mean	sklearn_hgb
E6_HYB_XGB_QUANTILES	hybrid	yes	yes	XGB-labelled	ensemble quantiles	sklearn_hgb
E7_HYB_GRU_MEAN	sequential hybrid	yes	yes	GRU-labelled	mean sequence	fallback MLP
E8_HYB_GRU_QUANTILES	sequential hybrid	yes	yes	GRU-labelled	quantile sequence	fallback MLP

(E4–E3), the effect of hybridization (E5–E3 and E6–E3), and the value of quantile descriptors relative to mean descriptors (E6–E5 and E8–E7). The same validation samples are used in every contrast.

2.9. Relation to prior stochastic physical–ML hybridization

The idea of feeding descriptors from a stochastic physical ensemble into an ML emulator is closely related to the hybridization strategy of Houénafa et al. (2025). The present study is framed as an ablation and audit extension of that idea. Rather than reporting only the best hybrid configuration, it decomposes the pipeline into physical memory, stochastic perturbation, ML correction and descriptor choice. It also evaluates whether the same ensemble that improves ML features can serve as a calibrated predictive uncertainty band. This distinction is important because physically plausible ensemble spread does not necessarily imply reliable probabilistic coverage (Atger 1999, Franz & Hogue 2011, Papacharalampous & Tyralis 2022).

2.10. Evaluation metrics and audit protocol

Point-prediction performance is evaluated with Nash–Sutcliffe efficiency (NSE), Kling–Gupta efficiency (KGE), root mean squared error (RMSE), mean absolute error (MAE), percent bias (PBIAS) and the coefficient of determination (R^2). Regime-specific metrics are computed for low, mid and high flows using thresholds estimated from training observations only. Low-flow NSE is interpreted cautiously because efficiency scores based on squared error are sensitive to low observed variance and can penalize dry-season errors in ways that are not proportional to their hydrological relevance (Pushpalatha et al. 2012).

Stochastic reference bands are evaluated using prediction interval coverage probability (PICP), prediction interval normalized average width (PINAW), mean prediction interval width (MPIW) and Winkler score. The intended behaviour is high coverage with the narrowest possible intervals, but sharpness is not interpreted as skill when coverage is far below the nominal level (Franz & Hogue 2011, Abaza et al. 2017, Papacharalampous & Tyralis 2022). The final audit checks chronological split integrity, feature construction, target leakage, backend metadata, common-window consistency and non-negative post-processing. The audit results are reported in Table 7.

3. Results

3.1. Common-window ranking

All experiments were evaluated on the same validation window defined in Section 2.2. Table 2 and Fig. 2 show a clear separation between the physical-only baselines, the pure ML baseline and the hybrid configurations. The deterministic and stochastic physical baselines (E0–E3) clustered within a narrow NSE range of 0.414–0.438, whereas the pure forcing-based ML model (E4) reached $NSE = 0.575$. All hybrid models exceeded E4 in NSE, indicating that the stochastic physical ensemble supplied additional predictive information beyond the observed forcing variables alone.

The strongest *backend-consistent* configuration was E6_HYB_XGB_QUANTILES, which achieved $NSE = 0.740$, $KGE = 0.810$, $RMSE = 0.275 \text{ m}^3 \text{ s}^{-1}$, $MAE = 0.162 \text{ m}^3 \text{ s}^{-1}$ and $PBIAS = 0.927\%$. This run is the main evidence for the proposed quantile-based stochastic physical–ML hybridization because its manuscript label and executed backend are consistent. The numerically best run was E8_HYB_GRU_QUANTILES ($NSE = 0.849$, $KGE = 0.899$ and

Table 2

Final common-window ranking. Metrics are computed on 4367 validation samples from 8 January 2009 to 31 December 2020.

Rank	Experiment	NSE	KGE	RMSE	MAE	PBIAS	Backend
1	E8_HYB_GRU_QUANTILES	0.849	0.899	0.210	0.124	1.579	fallback_mlp_on_flattened_sequences
2	E6_HYB_XGB_QUANTILES	0.740	0.810	0.275	0.162	0.927	sklearn_hgb
3	E7_HYB_GRU_MEAN	0.686	0.789	0.302	0.192	1.644	fallback_mlp_on_flattened_sequences
4	E5_HYB_XGB_MEAN	0.626	0.715	0.330	0.210	2.007	sklearn_hgb
5	E4_ML_PURE_XGB	0.575	0.661	0.352	0.223	0.742	sklearn_hgb
6	E1_RAMIS_DET_BF	0.438	0.714	0.405	0.248	0.860	physical
7	E3_RAMIS_LEVY_BF	0.429	0.725	0.408	0.255	-1.218	physical
8	E2_RAMIS_LEVY_NOBF	0.427	0.717	0.409	0.268	-7.551	physical
9	E0_RAMIS_DET_NOBF	0.414	0.713	0.413	0.270	-5.865	physical

RMSE = $0.210 \text{ m}^3 \text{ s}^{-1}$), but it is interpreted separately because the audit showed that it ran as a fallback multilayer perceptron on flattened sequences rather than as a recurrent GRU. The ranking is therefore reported in two layers: E8 is the best numerical configuration, whereas E6 is the strongest robust configuration for defensible process-based interpretation.

3.2. Hydrograph and flow-duration behaviour

The validation hydrograph in Fig. 3 shows that the hybrid models reproduce the timing and amplitude of seasonal streamflow fluctuations more closely than the physical-only baselines. The physical models retain the main wet-season response but frequently collapse towards near-zero discharge during recession periods. This behaviour is also evident in the flow-duration curves (Fig. 4), where E0 shows an unrealistic decline in the high-exceedance tail. Adding the baseflow reservoir reduces this problem only partially, indicating that the single linear storage term improves recession persistence but does not fully resolve dry-season memory.

The hybrid configurations provide a closer match to the observed flow-duration curve over most exceedance probabilities. The largest visual improvement occurs in the lower-flow tail, where quantile-based hybrids avoid the extreme collapse of the deterministic physical trajectory. Nevertheless, the hydrograph also shows remaining errors during sharp wet-season peaks. Thus, the hybrid improvement is not a complete correction of event-scale dynamics; it is mainly a reduction of systematic distributional errors and dry-season underestimation.

3.3. Ablation effects

Table 3 and Fig. 5 isolate the contribution of each modelling component. Adding the baseflow reservoir to the deterministic physical model increased NSE by 0.024 and reduced RMSE by $0.009 \text{ m}^3 \text{ s}^{-1}$ (E1–E0). Under stochastic forcing, the same structural addition produced a smaller but still positive gain in NSE and KGE (E3–E2). In contrast, adding Lévy-type perturbations to the baseflow-enabled physical model did not improve point-prediction skill: NSE decreased by 0.010 and RMSE increased by $0.004 \text{ m}^3 \text{ s}^{-1}$, although KGE increased slightly. This confirms that stochastic perturbation is not, by itself, the main source of point-prediction improvement.

The dominant gain came from hybridization. Relative to the complete stochastic physical baseline (E3), the pure forcing-based ML model (E4) improved NSE by 0.147, whereas the XGB-labelled hybrid using ensemble means (E5) improved NSE by 0.198. The quantile-based hybrid (E6) produced the largest backend-consistent ablation effect, with $\Delta\text{NSE} = 0.311$ and $\Delta\text{RMSE} = -0.133 \text{ m}^3 \text{ s}^{-1}$ relative to E3. The direct comparison between E6 and E5 added 0.114 NSE and 0.095 KGE, showing that ensemble quantiles carry useful predictive information that is lost when the stochastic physical ensemble is compressed to its mean alone.

The GRU-labelled contrasts must be interpreted with the backend audit in mind. E7 and E8 improved over their XGB-labelled counterparts, but these gains cannot be attributed to recurrent-network memory because both runs used a fallback MLP backend. They are therefore reported as additional sequential-feature experiments under fallback execution, not as evidence that GRU architecture is superior.

Table 3

Ablation effects on the common validation window. Positive Δ NSE and Δ KGE indicate improvement; negative Δ RMSE and Δ MAE indicate error reduction.

Ablation	Baseline	Candidate	Δ NSE	Δ KGE	Δ RMSE
deterministic_baseflow	E0_RAMIS_DET_NOBF	E1_RAMIS_DET_BF	0.024	0.000	-0.009
stochastic_baseflow	E2_RAMIS_LEVY_NOBF	E3_RAMIS_LEVY_BF	0.002	0.008	-0.003
levy_given_baseflow	E1_RAMIS_DET_BF	E3_RAMIS_LEVY_BF	-0.010	0.011	0.004
pure_ml_vs_physical	E3_RAMIS_LEVY_BF	E4_ML_PURE_XGB	0.147	-0.064	-0.056
hybrid_mean_vs_physical	E3_RAMIS_LEVY_BF	E5_HYB_XGB_MEAN	0.198	-0.010	-0.078
hybrid_quantiles_vs_physical	E3_RAMIS_LEVY_BF	E6_HYB_XGB_QUANTILES	0.311	0.085	-0.133
quantiles_vs_mean_xgb	E5_HYB_XGB_MEAN	E6_HYB_XGB_QUANTILES	0.114	0.095	-0.053
gru_mean_vs_xgb_mean	E5_HYB_XGB_MEAN	E7_HYB_GRU_MEAN	0.060	0.074	-0.028
gru_quantiles_vs_xgb_quantiles	E6_HYB_XGB_QUANTILES	E8_HYB_GRU_QUANTILES	0.109	0.089	-0.066

Table 4

RMSE ($\text{m}^3 \text{s}^{-1}$) by flow regime for representative models. Regime thresholds were estimated from calibration observations only.

Experiment	Low	Mid	High
E0_RAMIS_DET_NOBF	0.177	0.392	0.615
E1_RAMIS_DET_BF	0.197	0.365	0.613
E3_RAMIS_LEVY_BF	0.185	0.387	0.604
E4_ML_PURE_XGB	0.189	0.276	0.566
E6_HYB_XGB_QUANTILES	0.075	0.222	0.461
E8_HYB_GRU_QUANTILES	0.047	0.171	0.352

3.4. Performance by flow regime

Regime-specific RMSE values (Table 4 and Fig. 6) show that model skill is strongly flow-dependent. The physical-only configurations had similar errors across the selected regimes, with the largest absolute errors in the high-flow class. The pure ML model reduced mid-flow RMSE relative to the physical baselines, but its low-flow RMSE remained close to that of the physical models. The quantile-based hybrid E6 substantially reduced errors in all regimes, and the numerically best E8 configuration further reduced RMSE under fallback-MLP execution.

The largest proportional improvement occurred in the low-flow regime: RMSE decreased from $0.177 \text{ m}^3 \text{s}^{-1}$ for E0 to $0.075 \text{ m}^3 \text{s}^{-1}$ for E6 and $0.047 \text{ m}^3 \text{s}^{-1}$ for E8. The mid-flow and high-flow regimes also improved, but larger residual errors persisted during high flows, where event timing and peak magnitude remain more difficult to reproduce. These results support the interpretation from the flow-duration curves: stochastic ensemble quantiles are most useful for correcting systematic distributional errors, especially during recession and low-flow periods, while peak simulation remains only partially corrected.

3.5. Uncertainty diagnostics and backend audit

The stochastic physical intervals were strongly underdispersive (Table 5 and Fig. 7). E2 achieved PICP = 0.160 and E3 achieved PICP = 0.004, both far below the nominal 0.95 target. The corresponding PINAW values were also small, particularly for E3 (PINAW = 0.003), indicating that the intervals were narrow rather than reliably calibrated. The high Winkler scores confirm that most observed values fell outside the nominal intervals. Consequently, the stochastic ensemble should not be reported as a calibrated predictive uncertainty product in its current form. Its main demonstrated value is as a source of physically structured features for ML correction.

Table 6 summarizes the backend audit. The XGB-labelled experiments (E4–E6) used the `sklearn_hgb` backend, whereas the GRU-labelled experiments (E7 and E8) used `fallback_mlp_on_flattened_sequences`. This distinction changes the interpretation of the leaderboard: E8 may be reported as the best numerical experiment, but not as evidence for recurrent neural networks. The reproducibility checklist in Table 7 returned no FAIL item. WARN items were associated with missing observed-flow values, low-flow physical limitations, optional figure entries and the fallback backend, rather than with direct evidence of temporal leakage.

Table 5

Uncertainty diagnostics for stochastic physical reference bands.

Experiment	PICP	PINAW	MPIW	Winkler
E2_RAMIS_LEVY_NOBF	0.160	0.084	0.219	7.297
E3_RAMIS_LEVY_BF	0.004	0.003	0.009	10.050

Table 6

Backend-aware manuscript notes for ML and hybrid configurations.

Experiment	ML label	Backend label	Feature source	Manuscript claim
E4_ML_PURE_XGB	xgboost	sklearn_hgb	observed_forcing	report as gradient boosting (HGB)
E6_HYB_XGB_QUANTILES	xgboost	sklearn_hgb	stochastic_ensemble	report as gradient boosting (HGB)
E7_HYB_GRU_MEAN	gru	fallback_mlp_on_flattened_sequences	stochastic_ensemble	fallback MLP, not real recurrent GRU
E8_HYB_GRU_QUANTILES	gru	fallback_mlp_on_flattened_sequences	stochastic_ensemble	fallback MLP, not real recurrent GRU

Table 7

Reproducibility checklist for the current evidence package.

Item	Status	Evidence
Common validation intersection	PASS	leaderboard_common_intersection.csv
Anti-leakage audit	WARN	leakage_audit_status.txt=WARN
Physical audit	WARN	physical_audit_status.txt=WARN
Figure manifest	PASS	outputs/figures/phase34/phase34_figure_manifest.csv
GRU wording	WARN	E7/E8 use fallback backend
Optional physical components in figures	WARN	skipped entries in phase34_figure_manifest.csv

4. Discussion

4.1. Main inference from the ablation

The ablation results indicate that the principal source of improvement is not stochastic perturbation alone, nor the nominal complexity of the ML backend, but the transfer of distributional information from the stochastic physical ensemble to the emulator. This distinction is important for interpreting the experiment hierarchy. The deterministic and stochastic physical baselines remain within a narrow performance range, and direct Lévy perturbation of the baseflow-enabled physical model does not improve point-prediction skill. By contrast, the backend-consistent quantile hybrid E6 improves NSE by 0.311 relative to the full stochastic physical baseline E3 and by 0.165 relative to the pure forcing-based ML baseline E4. The most direct descriptor ablation, E6–E5, adds 0.114 NSE by replacing an ensemble-mean feature with a vector of ensemble quantiles while preserving the same executed backend. This result supports a specific interpretation of hybridization: the physical model is most useful when it is used to construct informative state-dependent predictors, rather than when its stochastic output is treated as a calibrated forecast distribution.

This interpretation is consistent with recent reviews of hydrological ML and hybrid modelling, which emphasize that performance gains can come either from more flexible algorithms or from more informative predictors (Papacharalampous & Tyralis 2022, Zhong et al. 2023). In StoHyMoLAP, the ablation evidence points mainly to the second mechanism. The stochastic physical ensemble transforms precipitation, evapotranspiration and antecedent basin conditions into a compact set of hydrologically filtered descriptors. The ML model then learns a residual or corrective mapping from these descriptors to observed discharge. The result is therefore not evidence that ML replaces hydrological structure; rather, it shows that a simple physical structure can become more useful when its distributional response is exposed to the emulator (Mekonnen et al. 2015, Houénafa et al. 2025).

4.2. Hydrological memory, baseflow and low-flow behaviour

The E1–E0 contrast shows that adding a linear baseflow reservoir produces a small but positive global gain in the deterministic physical baseline. This is hydrologically plausible for an Altiplano basin with strong wet–dry seasonality, delayed drainage and low-flow persistence. Previous Andean and mountain-catchment studies have similarly shown that groundwater storage, snow/glacier contributions, wetlands and season-dependent parameter behaviour can control dry-season discharge and recession dynamics (Nauditt et al. 2017, Ragettli et al. 2014, Condom et al. 2004, Lima-Quispe et al. 2025). The baseflow store in StoHyMoLAP should therefore be interpreted as a minimal representation of hydrological memory rather than as a complete description of subsurface storage.

The regime-specific results also show the limitation of this minimal representation. Although E1 improves the global deterministic score, the low-flow RMSE in Table 4 is not reduced relative to E0. The aggregate gain is therefore driven mainly by improvements in other portions of the hydrograph, whereas dry-season simulation remains structurally difficult. This is consistent with recession theory: a single linear reservoir cannot reproduce the effects of heterogeneous hillslope storage, wetlands, aquifer connectivity and spatially variable release times (Harman et al. 2009, van Esse et al. 2013). It also helps explain why the most substantial low-flow improvement appears only after hybrid correction with ensemble quantiles. In other words, the baseflow module adds physically meaningful memory, but it does not by itself solve the low-flow problem.

This result has implications for calibration. A global NSE objective is known to prioritize high-flow variance and can hide errors in the lower part of the flow-duration curve (Pushpalatha et al. 2012). Future versions should therefore include explicit low-flow criteria, such as inverse-flow NSE or regime-weighted losses, rather than relying only on global efficiency metrics. The broader ML literature on rainfall–runoff prediction also supports the importance of antecedent runoff or memory features for improving short-horizon runoff estimates (Adnan et al. 2021, Okkan et al. 2021). For applications where antecedent observed discharge is unavailable, physically simulated memory descriptors may offer a practical alternative, but their value must be tested against low-flow-specific metrics.

4.3. Lévy perturbations: useful ensemble diversity, not calibrated uncertainty

The Lévy component should be interpreted cautiously. Heavy-tailed perturbations are conceptually attractive in hydroclimatic settings where rainfall intermittency, extreme events and nonlinear runoff responses generate asymmetric response distributions. However, the current results show that adding Lévy perturbations to the physical model does not improve deterministic point-prediction skill. The E3–E1 contrast slightly reduces NSE and increases RMSE, despite a small gain in KGE. The perturbations therefore do not function as a standalone point-prediction improvement mechanism in the current configuration.

The uncertainty diagnostics are more restrictive. The nominal physical bands are strongly underdispersive, with PICP far below the nominal 0.95 target. This means that the ensemble spread is too narrow to be used as calibrated predictive uncertainty. In probabilistic hydrology, sharp intervals are valuable only when they are also reliable; scoring or reporting narrow bands without adequate coverage can be misleading (Atger 1999, Franz & Hogue 2011, Papacharalampous & Tyralis 2022). Similar underdispersion has been reported in hydrological ensemble prediction when forcing, parameter, structural and initial-state uncertainties are not represented jointly (Demirel, Booij & Hoekstra 2013, Thibault, Anctil & Boucher 2016, Crochemore, Ramos & Pappenberger 2016). StoHyMoLAP currently perturbs the physical response but does not yet provide a full uncertainty propagation and post-processing chain.

The negative uncertainty result does not make the stochastic ensemble irrelevant. It changes the role assigned to it. The ensemble is not yet a reliable uncertainty product, but its quantiles remain informative features for deterministic or conditional prediction. This distinction is central to the paper: stochastic simulation can fail as a calibrated interval while still creating useful descriptors of physical-response asymmetry, spread and tail behaviour. The hydrological value of the Lévy component is therefore mediated by the ML hybrid layer, not by direct interval coverage.

4.4. Why ensemble quantiles improve the hybrid model

The advantage of quantiles over ensemble means is the strongest and most defensible empirical result. An ensemble mean compresses all Monte Carlo members into a single central trajectory. This may be adequate when the conditional response distribution is approximately symmetric and homoscedastic, but it discards information about state-dependent spread, skewness and tail behaviour. In the Ramis basin, these properties are expected to vary between wet-season peaks and dry-season recession. Quantile features retain this variation by representing several points of the conditional distribution rather than only its first moment.

This mechanism also explains why the quantile hybrid outperforms the pure forcing-based ML baseline. Observed forcings describe atmospheric inputs, but they do not directly encode the catchment response state produced by the rainfall–runoff transformation. The physical ensemble acts as a hydrological feature generator: it converts meteorological forcing into discharge-consistent descriptors before ML correction. Similar physical-output enrichment strategies have been reported in hybrid hydrological modelling, where process-model outputs or synthetic physical simulations provide state information that is difficult for a data-driven model to infer from raw forcings alone (Mekonnen et al. 2015, Yang et al. 2020, Xie et al. 2021, Li et al. 2024, Zhu et al. 2025). The present results extend that idea by showing that quantiles of the stochastic physical response are more informative than the ensemble mean alone.

Nevertheless, the interpretation should remain conservative. The E6–E4 comparison combines two differences: the use of physical-ensemble features and the use of distributional descriptors. It does not by itself prove that all of the gain comes from physical structure rather than from temporal aggregation or feature engineering. A useful future ablation would compare ensemble quantiles against purely statistical lag quantiles derived from observed forcings or from simple persistence features. Such an experiment would separate the value of physical simulation from the value of quantile summarization.

4.5. Backend-aware interpretation of the sequential experiments

The numerical ranking places E8 first, but the backend audit prevents a standard recurrent-network interpretation. E7 and E8 were labelled as GRU experiments, yet the executed backend was a fallback multilayer perceptron applied to flattened sequences. These experiments therefore do not test the gating and hidden-state dynamics that define recurrent architectures. They should be described as GRU-labelled sequential hybrids under fallback-MLP execution, not as evidence that GRUs or other recurrent neural networks outperform gradient boosting (Kratzert et al. 2018, 2019).

The audit-adjusted interpretation is still useful. Quantile descriptors outperform mean descriptors in both executed emulator families: E6 exceeds E5 in the gradient-boosting family, and E8 exceeds E7 in the fallback sequential-feature family. This repeated ordering suggests that descriptor choice is more important than the nominal architecture label in the present evidence package. However, architecture-agnostic performance should be treated as a hypothesis, not as a final claim. A publication-ready extension should rerun E7 and E8 with an actual TensorFlow or PyTorch recurrent backend and compare the resulting architecture effect with the descriptor effect already measured here.

4.6. Reproducibility, leakage control and submission readiness

The audit tables strengthen the evidential value of the ablation by making the modelling workflow inspectable. The current evidence package reports strict temporal splitting, non-negative lags, sequence construction within temporal subsets, train-only scalers and train-only regime thresholds. These controls address common sources of optimistic performance in hydrological ML, especially when temporal dependence and lagged predictors are present. The audit therefore supports the main comparative finding: the improvement of E6 over E5 and E4 is unlikely to be an artefact of obvious temporal leakage.

At the same time, the WARN status should remain visible. WARN does not mean that leakage was detected, but it does identify issues that affect scientific interpretation: missing observed-flow values, low-flow physical limitations, underdispersive stochastic intervals, optional figure entries and fallback ML backends. Keeping these warnings in the main manuscript or supplementary material is preferable to hiding them, because it clarifies which conclusions are robust and which require additional computation. This position is aligned with calls for more transparent, reproducible and auditable hydrological modelling (Hall et al. 2021, Lees et al. 2021, Arsenault et al. 2022).

For submission to *Journal of Hydrology*, the strongest claim is therefore deliberately narrow: quantile descriptors extracted from a stochastic physical ensemble improve backend-consistent hybrid simulation of daily discharge in the Ramis basin. Claims about calibrated uncertainty, recurrent-network superiority or operational transferability should remain conditional until the uncertainty bands are recalibrated, the true recurrent backend is rerun and the final data provenance is fully documented.

5. Limitations

Five limitations define the scope of the present evidence. First, the experiments labelled as GRU were not executed with a true recurrent backend; the audit identified fallback multilayer-perceptron execution on flattened sequences. These runs are therefore useful as sequentially labelled stress tests, but they cannot support claims about recurrent

neural networks or long short-term memory learning in rainfall–runoff modelling (Kratzert et al. 2018). Second, the stochastic ensemble is not yet a calibrated predictive-uncertainty product. Its empirical coverage is far below the nominal 95% level, so its current value is as a source of distributional descriptors for ML correction rather than as an operational uncertainty band. Coverage recalibration, bias-aware post-processing and formal reliability assessment would be required before operational use (Thibault et al. 2016).

Third, the low-flow problem remains only partially resolved. The linear baseflow reservoir improves the global deterministic baseline, but the regime diagnostics show that it does not remove low-flow RMSE and bias. Future calibration should therefore include objective functions that explicitly weight low flows, such as inverse-flow criteria, rather than relying only on global NSE, KGE or RMSE (Pushpalatha et al. 2012). Fourth, the physical component is lumped and was evaluated in one strongly seasonal high-Andean basin. The conclusions should therefore be read as evidence for this hydroclimatic and modelling context, not as a claim of universal superiority over distributed hydrological models or over ML-only approaches in all basins. Fifth, the audit status is reported as WARN rather than fully PASS because missing observations, backend fallbacks and physical-audit warnings remain part of the current evidence chain. Final submission should preserve these audit records and should clearly state the definitive data source, station, units, preprocessing decisions and calibration settings used to reproduce the reported experiments.

6. Conclusions

This study evaluated StoHyMoLAP, an audit-controlled stochastic physical–machine-learning framework for daily streamflow simulation in the high-Andean Ramis basin. The central finding is that the main source of robust improvement is not the stochastic perturbation itself or the nominal complexity of the ML backend, but the use of quantile descriptors from the stochastic physical ensemble as features for the emulator.

The strongest backend-consistent hybrid, E6_HYB_XGB_QUANTILES, reached $NSE = 0.740$ and $KGE = 0.810$ over the common validation period. This corresponds to a gain of 0.311 NSE relative to the full stochastic physical baseline and 0.165 NSE relative to the pure forcing-based ML baseline. Replacing ensemble-mean features with ensemble quantiles alone added 0.114 NSE, confirming that the distributional structure of the physical ensemble carries useful information beyond a single mean trajectory.

The deterministic physical modifications had more limited effects. The linear baseflow reservoir produced a modest improvement in global skill, consistent with the need to represent slow catchment memory, but it did not fully correct low-flow errors. Direct Lévy perturbations did not reliably improve point-prediction skill when used only inside the physical simulator. Their strongest contribution was indirect: they created ensemble spread from which informative quantile descriptors could be extracted for ML correction.

The uncertainty analysis shows that predictive skill and uncertainty reliability must be assessed separately. The stochastic physical bands were strongly underdispersive, with PICP = 0.004 against a nominal 0.95 target. Thus, the present ensemble should not be interpreted as a calibrated uncertainty forecast. It is currently better understood as a feature-generation mechanism whose uncertainty representation requires additional recalibration before operational use.

The backend audit changes the interpretation of the best numerical score. E8_HYB_GRU_QUANTILES achieved the highest NSE, but it was executed as a fallback multilayer perceptron rather than as a true GRU. Consequently, the defensible conclusion is not that recurrent networks outperform the other models, but that quantile-based stochastic physical descriptors consistently improve hybrid streamflow simulation across the tested emulator families. Future work should rerun the sequential experiments with verified recurrent backends, recalibrate the stochastic ensemble for nominal coverage, incorporate low-flow-weighted objectives and test the framework across additional Altiplano basins.

Code and data availability

The manuscript tables and figures are generated from the StoHyMoLAP Phase 3.4B–3.5 result package. The scripts `make_method_diagram.py` and `build_paper_assets.py` reproduce all figures and CSV tables. Backend metadata should be regenerated after any change to the ML environment. The final submission should add the repository URL or DOI and the authoritative SENAMHI/ANA data-source citation once the public archive is assigned.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this draft, AI-assisted writing was used to organize the manuscript, convert reproducible artefacts into narrative text, and generate supporting LaTeX scripts. The authors remain responsible for verifying all scientific claims, numerical values, references, data provenance and final wording before submission.

CRedit authorship contribution statement

Jose Zevallos: Conceptualization of this study, Methodology, Software, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRedit authorship contribution statement

Jose Zevallos: Conceptualization of this study, Methodology, Software, Writing – original draft.

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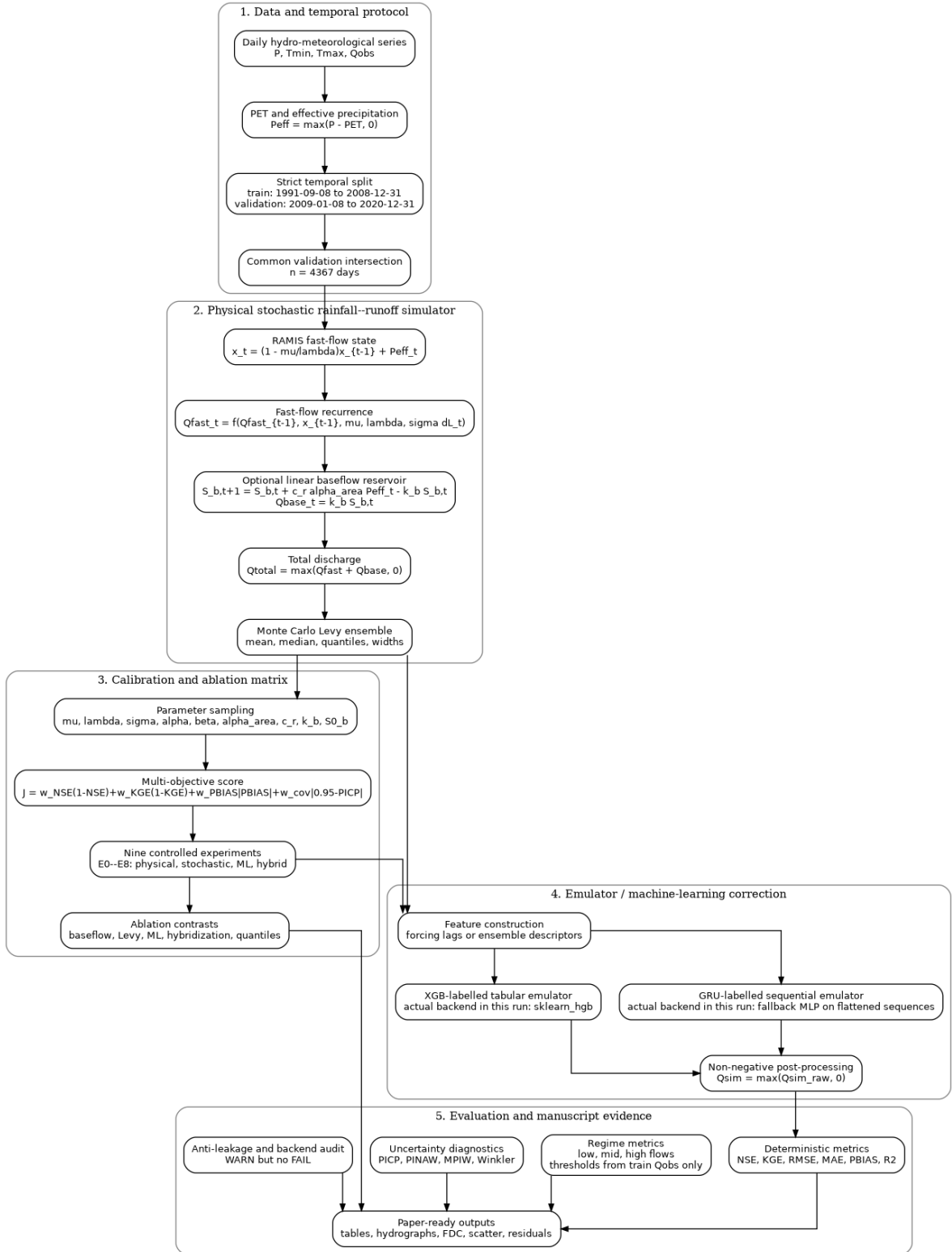


Figure 1: StoHyMoLAP workflow from daily hydro-meteorological data to lumped physical simulation, Lévy-type stochastic ensemble generation, ensemble-descriptor construction, ML emulation, ablation comparison and audit-controlled manuscript evidence.

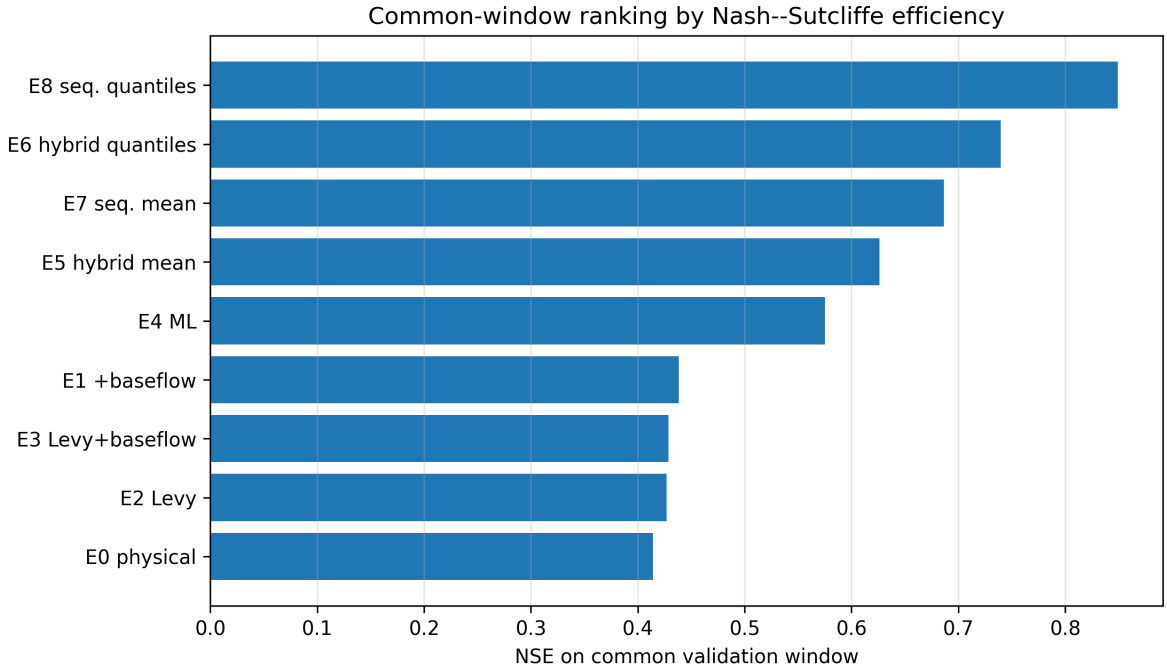


Figure 2: Ranking of all experiments by common-window NSE. Numerical values correspond to Table 2.

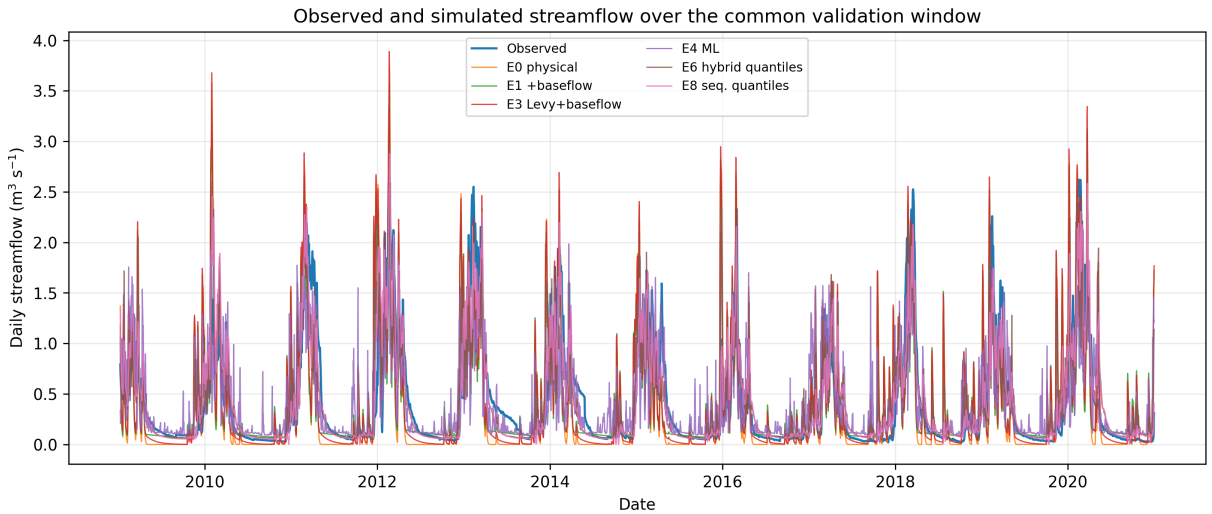


Figure 3: Observed and simulated daily discharge over the common validation window for representative physical, pure ML and hybrid models.

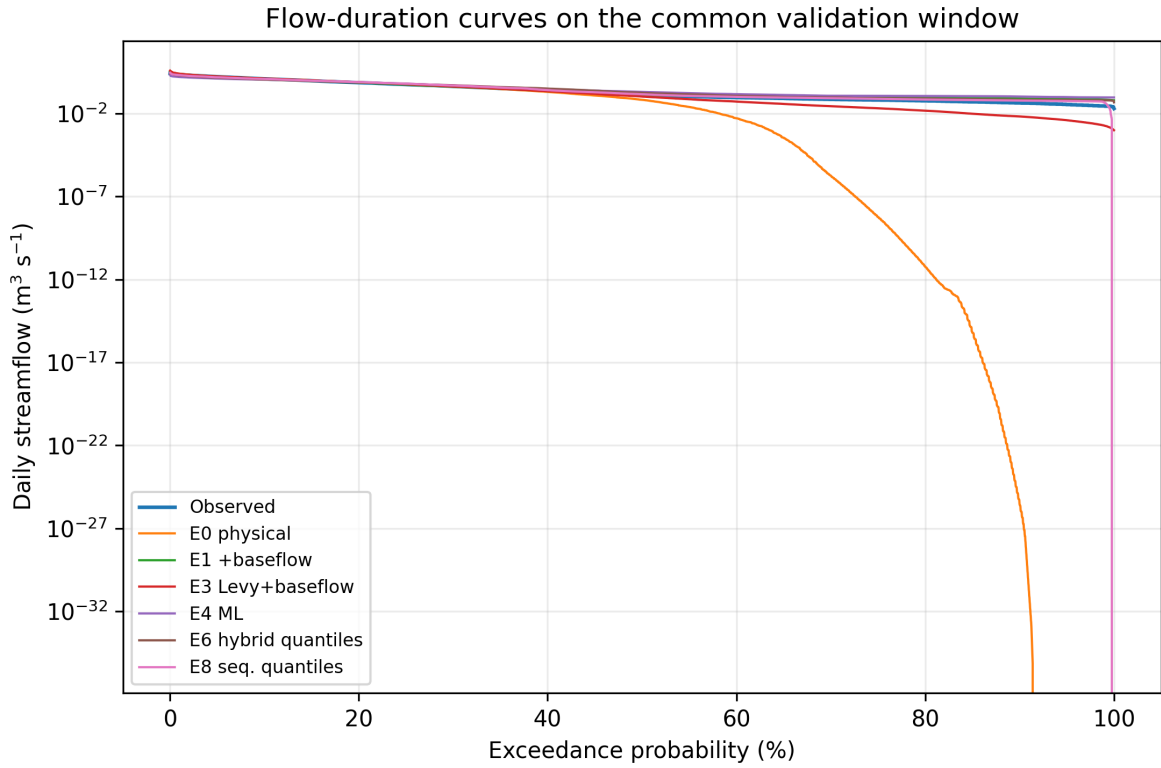


Figure 4: Flow-duration curves for observed discharge and representative simulations over the common validation window.

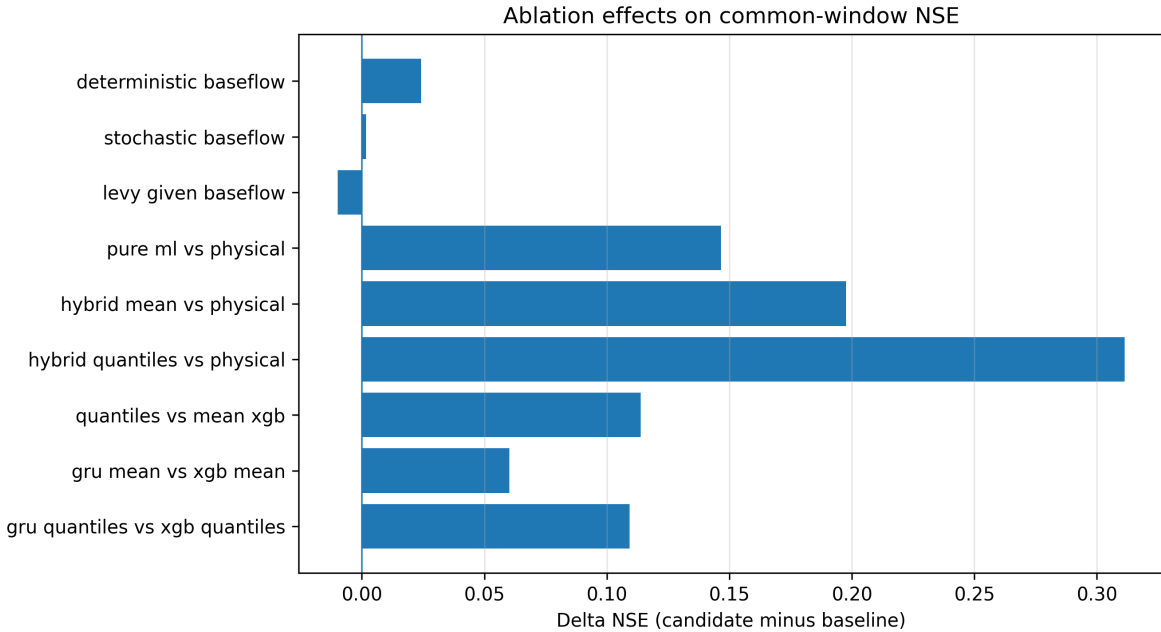


Figure 5: Ablation effects expressed as candidate-minus-baseline Δ NSE. Numerical values are reported in Table 3.

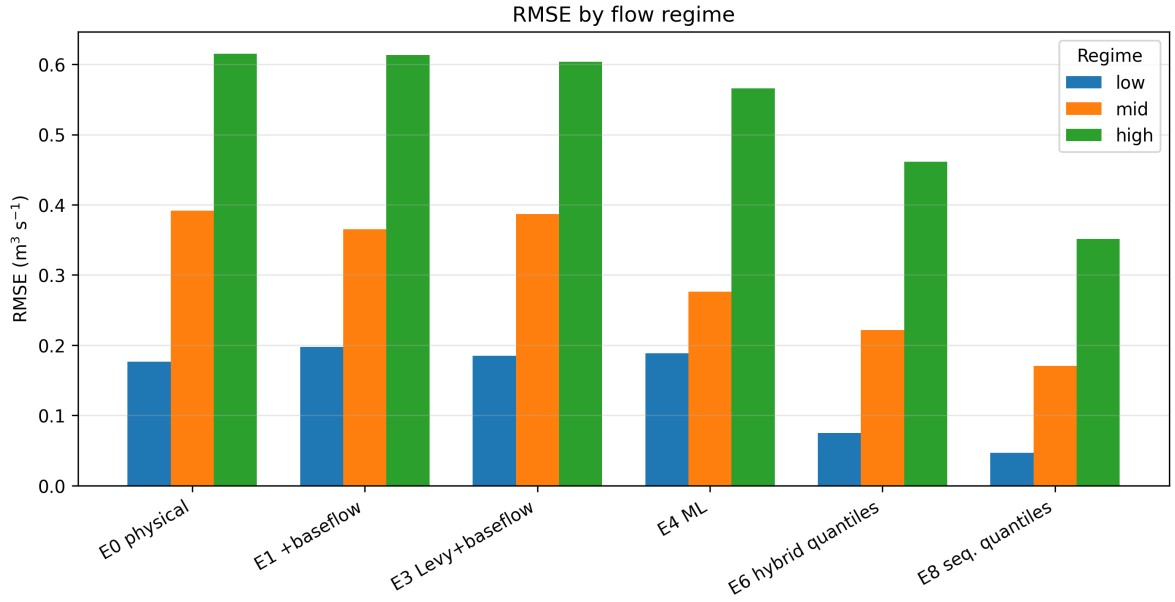


Figure 6: RMSE by low, mid and high flow regimes for representative models. Values correspond to Table 4.

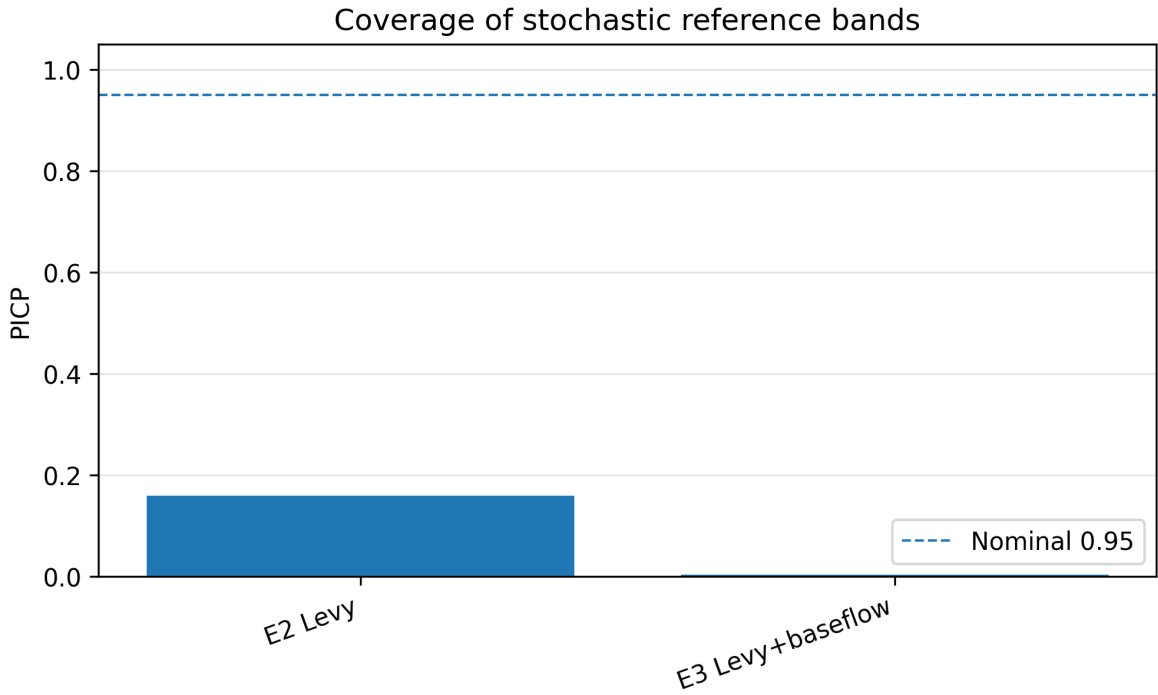


Figure 7: Prediction interval coverage probability for stochastic physical references. The dashed line marks the nominal 0.95 target.